

VOXGEST

Elaborated System Logic and Decision Notes

Defense-Ready Edition

A plain-English explanation of every major idea, approach, rejected option, relevance, irrelevance, and Chapter 1-3 logic behind the creation of VoxGest.

Field	Details
Project	VoxGest: A Bidirectional Digital Accessibility Application for Sign Language Recognition and Real-Time Speech Communication Using Deep Learning and Visual Tracking
Prepared for	Team review, adviser discussion, Chapter 1-3 defense preparation, and oral-defense logic alignment
Style	Very simple. Technical ideas are explained using ordinary language and clear decision cards.
Main truth	VoxGest is not only a CNN alphabet recognizer anymore. It is a layered Android accessibility prototype using landmarks, static recognition, dynamic recognition, NOTHING/no-output control, gates, token composition, speech, and avatar/fingerspelling feedback.

Read this first:

This document explains the logic behind the paper and system. It is not a replacement for the thesis. It is a companion guide so teammates can understand what to say, why the system changed, and how every part connects.

1. The Whole System in One Very Simple Story

Imagine one phone placed between two people. One person signs. The phone watches the hand and body. The phone does not immediately trust every prediction. It first checks whether the prediction is stable. If the prediction passes the checks, the app writes the word or letter, speaks it if needed, and can show a simple avatar or fingerspelling response. The hearing person can also speak back, and the app can show the spoken reply as text or visual feedback.

Donkey-simple truth:

VoxGest is a small communication bridge. It does not claim to replace a human interpreter. It helps with selected, controlled, common communication when no interpreter is around.

Part	Simple meaning	Why it exists
Camera	The eyes of the system.	It sees the signer without gloves or special sensors.
MediaPipe / Holistic	The dot maker.	It turns the hand and body into points that the AI can measure.
Features	Numbers from the dots.	AI models understand numbers better than raw messy camera images.
Model	The trained guesser.	It learned examples of signs and predicts what the new sign looks like.
Gates	The security guard.	They block shaky, weak, fake, or lowconfidence predictions.
NOTHING	The quiet label.	It teaches the model that sometimes the correct output is no output.
Token composer	The sentence builder.	It builds text only from accepted signs, not raw guesses.
TTS/STT	The mouth and ears of the app.	TTS speaks signed text; STT captures the hearing person reply.

Avatar/fingerspelling	The visual helper.	It gives the signing user a visual response path.
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2. What Chapter 1, Chapter 2, and Chapter 3 Must Prove

Chapter	What it must prove	Simple explanation
Chapter 1	There is a real communication problem and VoxGest has a realistic purpose.	This chapter answers: Who has the problem? Why does it matter? What are we building? What are the limits?
Chapter 2	The literature supports the exact system choices.	This chapter answers: Why landmarks? Why deep learning? Why sequence models? Why Android/offline? Why ISO/IEC 25010?
Chapter 3	The method matches the real system, not an old idea.	This chapter answers: How did we build it? How did the models change? How will we test and evaluate it?

Big rule:

The paper, demo, and source files must tell one story. If Chapter 1 says CNN-only but the demo uses landmarks and TCN, the panel can easily question the team.

3. Full Logic Path of the Project

1. Problem: Signing users and hearing/non-signing people struggle to communicate when no interpreter is available.
2. Need: A phone-based tool is more available than a human interpreter in many daily situations.
3. First technical idea: Use CNN because CNN is common for image recognition.
4. Live problem: Raw images were affected by lighting, background, cropping, hand distance, and camera angle.
5. Better feature strategy: Use MediaPipe landmarks so the model focuses on hand/body geometry instead of the whole room.
6. Static foundation: Use 63 hand features for A-Z, del, space, and nothing.
7. Practical speed-up: Add dynamic word recognition because spelling every word letter by letter is slow.
8. Motion requirement: Use 30-frame sequences because moving signs need time, not one still image.
9. Body-location requirement: Use pose + hand features because some signs depend on where the hand is near the face, chin, chest, or body.
10. Safety requirement: Add NOTHING and gates because the system must not speak random words from idle or partial movement.
11. Communication requirement: Add token composer so only accepted predictions become user-facing text.
12. Bidirectional requirement: Add speech-to-text and avatar/fingerspelling so the hearing person can reply in a visible way.
13. Defense requirement: Freeze phrase expansion and vocabulary expansion until the core alphabet, 10 words, and NOTHING are reliable.
14. Future research path: Test FullSign225 and Residual Dilated TCN separately, without overwriting the protected baseline.

4. Development Timeline: What Changed and Why

When	What happened	Why it mattered
March 30, 2026	Project started; environment and ASL alphabet data prepared.	Created the first dataset and machine learning starting point.
March 30-31	Initial CNN alphabet model was trained.	Proved that image classification could work under dataset conditions.
April 1-3	Live webcam testing exposed prediction, crop, and domain problems.	Showed that dataset validation is not the same as live readiness.
April 3-5	Scope expanded from one-way sign-to-speech to bidirectional communication.	A real communication bridge must allow both people to respond.
April 5-8	Hybrid CNN + landmark attempt was explored.	The team tested whether image features plus geometry could solve confusion.

April 8-10	Landmark-only static recognition became stronger.	Reduced sensitivity to background, lighting, and image crop problems.
April 22	NOTHING, gates, and 30x162 dynamic fixes were introduced.	Started reducing false word output and improving dynamic recognition control.
April 24	Holistic 162-feature dynamic recognition stabilized a smaller word set.	Solved part of the floating-hand/bodylocation problem.
April 28-29	Dominant-hand policy, single-hand pose masking, and mirror mapping clarified.	Aligned recording, training, testing, and Android preprocessing.
May 8	Phrase-intent pathway was created for ASK_NAME but required endpoint segmentation.	Showed that long phrases must not be treated like short 30-frame words.
May 10 onward	Recognition Hardening v1 began; phrase expansion frozen; token composer and avatar contract added.	Stability became more important than adding more features.
May 19-21	FullSign225 and team recorder workflow tested.	Created a separate experimental path for fuller two-hand/body information.

5. Approach Decision Cards

This section explains each major decision using the same pattern: what we did, why we used it, why not another approach, how it works, and how it connects to Chapters 1-3.

Decision 1: Build a phone-based accessibility app

Donkey-simple idea:

Use one phone like a small bridge between two people.

Question	Answer
WHAT we did	The team designed VoxGest as a mobile assistive communication prototype.
WHY we used it	A phone is common, portable, and practical in hospitals, schools, offices, homes, interviews, and public counters.
WHY NOT the other approach	A desktop-only system would not match the daily communication problem. A hardware glove would add cost and reduce accessibility.
HOW it works	The phone camera captures signs, the app processes the input, and output appears as text, speech, or visual response.
WHY it is relevant	Very relevant because Chapter 1 is about a real communication problem and a practical mobile solution.
WHAT is irrelevant or delayed	Specialized lab equipment is irrelevant for the current capstone goal because the system should work with ordinary devices.
CHAPTER 1-3 connection	Chapter 1 background and purpose; Chapter 3 Android architecture.

Decision 2: Use Android as the first deployment platform

Donkey-simple idea:

Python is the kitchen; Android is the restaurant where users receive the finished meal.

Question	Answer
WHAT we did	The target prototype is Android-based, with Kotlin, CameraX, MediaPipe-style landmarks, and TensorFlow Lite.
WHY we used it	Android phones are common, portable, and support camera, speech, and on-device ML tools.
WHY NOT the other approach	A web-only or desktop-only deployment would be easier for testing but less aligned with the portable accessibility goal.
HOW it works	Training happens in Python; exported model files and labels are consumed by Android.
WHY it is relevant	Relevant because the thesis title and objectives need a real mobile application direction.
WHAT is irrelevant or delayed	Training directly on Android is not needed now; Android should run models, not train them.
CHAPTER 1-3 connection	Chapter 1 scope; Chapter 3 Android Application Development.

Decision 3: Keep recognition offline and on-device

Donkey-simple idea:

The phone should think locally instead of asking the internet every time.

Question	Answer
WHAT we did	VoxGest is designed for on-device inference using TensorFlow Lite.
WHY we used it	Offline recognition supports privacy, lower latency, and use in places with weak internet.
WHY NOT the other approach	Cloud recognition may be powerful, but it creates privacy, connection, and delay concerns.
HOW it works	The app loads TFLite models and runs inference on the device.
WHY it is relevant	Relevant because accessibility tools should work in realistic environments, not only with strong Wi-Fi.
WHAT is irrelevant or delayed	Cloud backend recognition is delayed/irrelevant for the current defense-ready local prototype.
CHAPTER 1-3 connection	Chapter 1 project context; Chapter 2 edge computing; Chapter 3 runtime architecture.

Decision 4: Start with CNN image recognition

Donkey-simple idea:

At first, the AI looked at hand pictures like flashcards.

Question	Answer
WHAT we did	The project began with CNN because CNNs are common for visual classification.
WHY we used it	It was a logical first step for static ASL hand images and easy to explain academically.
WHY NOT the other approach	CNN alone was not enough because live camera images differ from clean training images.
HOW it works	Images were resized and trained through convolution layers to classify hand signs.
WHY it is relevant	Relevant as project history and RRL support for image-based recognition.
WHAT is irrelevant or delayed	CNN should not be presented as the final main dynamic recognizer.
CHAPTER 1-3 connection	Chapter 2 CNN literature; Chapter 3 model evolution.

Decision 5: Move away from CNN-only

Donkey-simple idea:

The whole picture was too messy, so the team started measuring useful dots instead.

Question	Answer
WHAT we did	The team shifted from raw image reliance to landmark-based features.
WHY we used it	Live camera testing showed background, lighting, cropping, hand distance, and angle problems.
WHY NOT the other approach	Staying CNN-only would make the paper and demo weak because webcam reliability was already a known issue.
HOW it works	CNN is now explained as an early attempt; landmarks became the current core feature strategy.
WHY it is relevant	Relevant because it makes the methodology honest and defense-ready.
WHAT is irrelevant or delayed	It is irrelevant to pretend that the final system is still only CNNbased.
CHAPTER 1-3 connection	Chapter 1 correction; Chapter 3 model development.

Decision 6: Use MediaPipe landmarks

Donkey-simple idea:

The system follows dots on the hand/body instead of staring at the whole room.

Question	Answer
WHAT we did	MediaPipe-style landmarks convert hands and body into x, y, z coordinates.
WHY we used it	Landmarks reduce background and lighting noise and make Python-to-Android preprocessing more controllable.
WHY NOT the other approach	Raw pixels contain too much irrelevant information like room color, skin lighting, and background objects.
HOW it works	The frame becomes numerical landmark features used by static or dynamic models.
WHY it is relevant	Highly relevant because this is the real bridge between camera and model input.
WHAT is irrelevant or delayed	Full-image recognition is less relevant for the current live mobile pipeline.
CHAPTER 1-3 connection	Chapter 2 landmark tracking; Chapter 3 feature extraction.

Decision 7: Use 63 features for static alphabet signs

Donkey-simple idea:

One hand has 21 dots, and every dot has x, y, z. $21 \times 3 = 63$.

Question	Answer
WHAT we did	Static alphabet/control signs use 63 normalized hand features.
WHY we used it	Static letters mostly need hand shape, not the whole body motion.
WHY NOT the other approach	Using full pose for static letters would add complexity without much benefit for simple hand-shape recognition.
HOW it works	Extract 21 hand landmarks, subtract wrist position, scale by hand size, and feed 63 numbers to the static model.
WHY it is relevant	Relevant because spelling remains the fallback when word recognition is weak.
WHAT is irrelevant or delayed	A 225-feature full-body input is unnecessary for simple static letters.
CHAPTER 1-3 connection	Chapter 1 scope; Chapter 3 static feature engineering.

Decision 8: Add dynamic word recognition

Donkey-simple idea:

Letters are small bricks; words are bigger blocks that make communication faster.

Question	Answer
WHAT we did	The system added selected dynamic words such as YES, NO, PLEASE, WATER, HELLO, HELP, STOP, DOCTOR, NAME, and THANKYOU.
WHY we used it	Spelling every word letter by letter is slow for real communication.
WHY NOT the other approach	Trying to recognize every possible ASL word is unrealistic for the current capstone stage.
HOW it works	The system uses temporal landmark sequences to recognize selected moving words.
WHY it is relevant	Relevant because it makes VoxGest more useful than alphabet spelling alone.
WHAT is irrelevant or delayed	Unlimited vocabulary is delayed until the current demo words become reliable.
CHAPTER 1-3 connection	Chapter 1 purpose/objectives; Chapter 3 dynamic recognition.

Decision 9: Use 30 frames for dynamic words

Donkey-simple idea:

A moving sign is like a short movie, not one photo.

Question	Answer
WHAT we did	Dynamic word signs are represented as 30-frame sequences.
WHY we used it	The model needs start, middle, and end movement to understand a word sign.

WHY NOT the other approach	A single-frame classifier cannot reliably understand motionbased signs.
HOW it works	Each sign attempt is sampled into a fixed 30-frame input window.
WHY it is relevant	Relevant because dynamic recognition must explain time, not only shape.
WHAT is irrelevant or delayed	Static image-only logic is irrelevant for moving words.
CHAPTER 1-3 connection	Chapter 3 dynamic feature engineering and model input contract.

Decision 10: Use 162 features for one-hand dynamic words

Donkey-simple idea:

The model needs to know if the hand is near the face, mouth, chin, or chest.

Question	Answer
WHAT we did	The demo10/onehand dynamic path uses 99 pose features plus 63 dominant-hand features per frame.
WHY we used it	Hand-only data caused a floating-hand problem because the model saw hand motion without body location.
WHY NOT the other approach	Using hand-only 63 features for dynamic words was not enough for body-referential signs.
HOW it works	Each frame contains pose and dominant-hand landmarks, usually anchored to body/face context.
WHY it is relevant	Relevant because it explains why body context was added without requiring a full two-hand default.
WHAT is irrelevant or delayed	Using only wrist-normalized hand coordinates is delayed/limited for dynamic signs that need body location.
CHAPTER 1-3 connection	Chapter 3 feature engineering; Chapter 2 dynamic recognition literature.

Decision 11: Keep a dominant-hand policy

Donkey-simple idea:

The system chooses a main hand so it does not get confused between left and right.

Question	Answer
WHAT we did	The current baseline prioritizes one configured or selected dominant hand.
WHY we used it	It supports one-hand accessibility and makes data collection, training, testing, and Android preprocessing consistent.
WHY NOT the other approach	Full two-hand recognition is more complex and can weaken a small dataset if used too early.
HOW it works	The pipeline selects the configured hand, applies single-hand pose policy, and masks irrelevant pose movement when needed.
WHY it is relevant	Relevant because the project has an accessibility reason and a reliability reason.
WHAT is irrelevant or delayed	Two-hand default is delayed for the protected baseline but explored separately in FullSign225.
CHAPTER 1-3 connection	Chapter 1 accessibility logic; Chapter 3 data and feature policy.

Decision 12: Create FullSign225 as a separate experiment

Donkey-simple idea:

Sometimes one hand is not enough, so a separate test watches pose plus both hands.

Question	Answer
WHAT we did	FullSign225 uses 99 pose + 63 left hand + 63 right hand = 225 features.
WHY we used it	Some signs may need two-hand or fuller body information, so this experiment tests whether more landmarks help.
WHY NOT the other approach	It should not replace demo10 automatically because experimental models can break a stable demo.

HOW it works	Teammates record landmark samples; the main PC audits, merges, trains, and tests one model.
WHY it is relevant	Relevant as a future strengthening path, especially for signs where both hands matter.
WHAT is irrelevant or delayed	It is irrelevant to make FullSign225 the default before live proof and comparison.
CHAPTER 1-3 connection	Chapter 3 future/experimental path; Chapter 1 limitation.

Decision 13: Use TCN as the main current dynamic direction

Donkey-simple idea:

TCN is a motion watcher for short fixed-length sign videos.

Question	Answer
WHAT we did	The current dynamic training direction uses TCN, with LSTM kept as comparison/back-up.
WHY we used it	TCN fits fixed 30-frame landmark sequences and is practical for TensorFlow Lite deployment.
WHY NOT the other approach	CNN is not suitable for current dynamic word recognition because dynamic signs are not just still images.
HOW it works	Temporal convolution learns movement patterns across the 30frame sequence.
WHY it is relevant	Relevant because it aligns with lightweight on-device recognition.
WHAT is irrelevant or delayed	Transformers/ST-GCN are delayed because they need more data, time, and careful engineering.
CHAPTER 1-3 connection	Chapter 2 sequential models; Chapter 3 model development.

Decision 14: Keep LSTM as comparison/back-up

Donkey-simple idea:

LSTM is another motion reader, but not the only answer.

Question	Answer
WHAT we did	LSTM remains part of the project as a sequence model comparison and earlier dynamic approach.
WHY we used it	It helped the team understand temporal recognition and remains useful for comparison.
WHY NOT the other approach	LSTM does not automatically fix bad landmarks, bad data, or weak live gates.
HOW it works	It reads frame sequences step by step and predicts a dynamic word.
WHY it is relevant	Relevant as part of model evolution and comparison.
WHAT is irrelevant or delayed	It is irrelevant to claim LSTM alone solves recognition reliability.
CHAPTER 1-3 connection	Chapter 3 model history; Chapter 2 dynamic recognition.

Decision 15: Add Residual Dilated TCN as a safe experiment

Donkey-simple idea:

A stronger TCN can see wider motion without becoming too huge.

Question	Answer
WHAT we did	Residual Dilated TCN is proposed as an experimental upgrade, separate from protected baseline files.
WHY we used it	It may improve signs where start-middle-end motion matters.
WHY NOT the other approach	It should not overwrite the working baseline because stronger models can still fail if data is dirty.
HOW it works	Dilation expands the time view; residual connections preserve earlier useful information.
WHY it is relevant	Relevant as a controlled improvement path.
WHAT is irrelevant or delayed	It is irrelevant to present RD-TCN as finished unless tested with live logs.
CHAPTER 1-3 connection	Chapter 3 experimental model path; future work/limitations.

Decision 16: Delay Transformer or ST-GCN

Donkey-simple idea:

A big truck is not needed if the road is still being built.

Question	Answer
WHAT we did	Transformer/ST-GCN are recognized as possible future research models, not current priority.
WHY we used it	They may be powerful, but they require more data, engineering time, and careful validation.
WHY NOT the other approach	Using them now can distract from fixing the current data, gates, Android contract, and live reliability.
HOW it works	They can be studied later after stable datasets and baselines exist.
WHY it is relevant	Relevant as future research, not current implementation.
WHAT is irrelevant or delayed	Irrelevant for the immediate Chapter 1-3 defense-ready scope.
CHAPTER 1-3 connection	Chapter 2 future literature; Chapter 3 limitation/future work.

Decision 17: Add NOTHING as a negative/no-output class

Donkey-simple idea:

NOTHING teaches the AI when to keep quiet.

Question	Answer
WHAT we did	The system includes NOTHING to represent idle, partial, transition, or non-sign movement.
WHY we used it	A classifier forced to choose a word will produce fake words from random movement.
WHY NOT the other approach	Without NOTHING, the app could speak or animate incorrect words, hurting trust.
HOW it works	Record idle hands, open hands, partial signs, aborted signs, and transitions as NOTHING.
WHY it is relevant	Highly relevant to reliability and safety.
WHAT is irrelevant or delayed	NOTHING is not a word to display, speak, or animate.
CHAPTER 1-3 connection	Chapter 1 reliability objective; Chapter 3 negative class testing.

Decision 18: Use recognition gates

Donkey-simple idea:

The security guard checks the model before letting a prediction enter the app.

Question	Answer
WHAT we did	The system checks confidence, margin, motion, wrist path, hand presence, quality, stability, and cooldown.
WHY we used it	Raw model predictions can be unstable; gates prevent weak guesses from becoming output.
WHY NOT the other approach	Letting raw predictions go directly to text/speech/avatar is unsafe and can embarrass the demo.
HOW it works	Only predictions that pass the rules become accepted tokens.
WHY it is relevant	Relevant because defense panels care about reliability, not only accuracy.
WHAT is irrelevant or delayed	A pure accuracy-only approach is incomplete for real-time communication.
CHAPTER 1-3 connection	Chapter 3 recognition hardening and testing.

Decision 19: Use token composer

Donkey-simple idea:

The model guesses; the composer builds the message only from approved pieces.

Question	Answer
WHAT we did	The token composer builds text from accepted letters, spaces, deletes, and word tokens.

WHY we used it	It separates raw prediction from final output, making the system safer and easier to control.
WHY NOT the other approach	Directly speaking every prediction would create false communication.
HOW it works	Accepted A-Z append letters; space commits separation; del deletes; accepted words append tokens; NOTHING is ignored.
WHY it is relevant	Relevant because it supports sentence output without claiming full ASL grammar.
WHAT is irrelevant or delayed	Raw prediction-to-speech is irrelevant for safe user-facing output.
CHAPTER 1-3 connection	Chapter 1 purpose; Chapter 3 output generation.

Decision 20: Freeze phrase-intent expansion for now

Donkey-simple idea:

Do not build the second floor while the first floor is still shaking.

Question	Answer
WHAT we did	Phrase recognition exists as an explored path but is disabled by default during Recognition Hardening v1.
WHY we used it	Phrase shortcuts need endpoint segmentation and stable basics first.
WHY NOT the other approach	Treating a long phrase like a short 30-frame word caused early/premature recognition.
HOW it works	Phrase intents should be classified only after a complete motion segment, such as 60 frames after final hold.
WHY it is relevant	Relevant as future work, but the current defense scope should focus on stable basics.
WHAT is irrelevant or delayed	Phrase expansion is irrelevant for the current hardening pass until alphabet, words, NOTHING, and token composer pass live tests.
CHAPTER 1-3 connection	Chapter 1 limitation; Chapter 3 future/optional pathway.

Decision 21: Use endpoint segmentation for phrases

Donkey-simple idea:

A phrase should be judged only after the whole gesture is finished.

Question	Answer
WHAT we did	The phrase path uses IDLE -> START -> ACTIVE -> END_HOLD -> CLASSIFY logic.
WHY we used it	Longer phrase motions cannot be safely classified from a rolling short window.
WHY NOT the other approach	Without endpoint segmentation, the system may output a phrase too early.
HOW it works	The system waits for a complete segment, resamples it, then classifies once.
WHY it is relevant	Relevant to explain why phrase-intent design exists but is controlled.
WHAT is irrelevant or delayed	Rolling 30-frame phrase output is irrelevant/unsafe for phrase gestures.
CHAPTER 1-3 connection	Chapter 3 phrase-intent methodology/future path.

Decision 22: Use speech-to-text and text-to-speech

Donkey-simple idea:

Both people need a way to send and receive messages.

Question	Answer
WHAT we did	The system includes TTS for signed output and STT for hearing-user replies.
WHY we used it	A one-way app helps only half of the conversation.
WHY NOT the other approach	Sign-to-text only is less useful because the hearing person still needs a way to reply visibly.

HOW it works	Recognized signed text can be spoken; spoken replies can become visible text.
WHY it is relevant	Relevant because the title says bidirectional communication.
WHAT is irrelevant or delayed	A one-direction-only scope is outdated for the current paper.
CHAPTER 1-3 connection	Chapter 1 purpose; Chapter 3 speech module.

Decision 23: Use lightweight avatar/fingerspelling

Donkey-simple idea:

The signing user also needs to see the reply, not just hear it.

Question	Answer
WHAT we did	Known words may play simple avatar animations; unknown words may fingerspell.
WHY we used it	The reverse path must be visible to a Deaf/signing user.
WHY NOT the other approach	A full ASL avatar is too large and risky for the current capstone stage.
HOW it works	Avatar playback happens only after token composition; NOTHING triggers no animation.
WHY it is relevant	Relevant because it completes the bidirectional concept at prototype level.
WHAT is irrelevant or delayed	Claiming full ASL grammar animation is irrelevant and inaccurate for now.
CHAPTER 1-3 connection	Chapter 1 bidirectional goal; Chapter 3 avatar module.

Decision 24: Train in Python, run in Android

Donkey-simple idea:

Python is for cooking the model; Android serves the cooked model to users.

Question	Answer
WHAT we did	The team trains and repairs models in Python, then exports TFLite assets for Android.
WHY we used it	Python provides flexible ML training tools, while Android provides the user-facing app.
WHY NOT the other approach	Training on Android is not required and would complicate the app.
HOW it works	Scripts collect data, train models, export labels/models; Android consumes them.
WHY it is relevant	Relevant because it separates engineering responsibilities clearly.
WHAT is irrelevant or delayed	Direct Android training is irrelevant for the current prototype.
CHAPTER 1-3 connection	Chapter 3 development workflow and Android handoff.

Decision 25: Use teammate recorder pack

Donkey-simple idea:

Everyone can collect ingredients, but one kitchen cooks the final meal.

Question	Answer
WHAT we did	Teammates record landmark samples; the main PC merges, audits, trains, and exports one final model.
WHY we used it	This avoids inconsistent environments, different models, and messy merging of trained files.
WHY NOT the other approach	Combining separately trained teammate models is not the correct workflow for the current setup.
HOW it works	Each teammate produces samples; the main PC checks counts and trains a shared model.
WHY it is relevant	Relevant because team data collection supports generalization without losing control.
WHAT is irrelevant or delayed	Individual teammate-trained models are irrelevant unless a formal federated approach is designed.
CHAPTER 1-3 connection	Chapter 3 dataset development and training control.

Decision 26: Use live testing as part of methodology

Donkey-simple idea:

A classroom score is not enough; the system must pass the camera test.

Question	Answer
WHAT we did	Live webcam/mobile testing is treated as a formal development and evidence step.
WHY we used it	Validation accuracy does not prove real camera reliability.
WHY NOT the other approach	Relying only on training accuracy can hide lighting, mirror, hand, and motion issues.
HOW it works	Use live logger, diagnostics, alphabet protocol, word tests, and failure reports.
WHY it is relevant	Relevant because the system is meant to be real-time.
WHAT is irrelevant or delayed	A paper-only or validation-only claim is irrelevant for defense readiness.
CHAPTER 1-3 connection	Chapter 3 testing procedure.

Decision 27: Use ISO/IEC 25010 evaluation

Donkey-simple idea:

Do not only ask: Did it guess? Ask: Is the app useful, fast, stable, and usable?

Question	Answer
WHAT we did	The evaluation uses functional suitability, performance efficiency, reliability, and usability.
WHY we used it	Capstone systems need software quality evaluation, not only model accuracy.
WHY NOT the other approach	Accuracy alone cannot measure user experience, interface clarity, latency, or reliability.
HOW it works	Use structured Likert-scale evaluation with IT professionals and target/representative users where appropriate.
WHY it is relevant	Relevant because it gives Chapter 4 a clear evaluation path.
WHAT is irrelevant or delayed	Unstructured opinions alone are not enough for formal evaluation.
CHAPTER 1-3 connection	Chapter 1 objectives; Chapter 3 evaluation method.

Decision 28: Keep honest limitations

Donkey-simple idea:

A good defense tells the truth before the panel attacks it.

Question	Answer
WHAT we did	The paper states that VoxGest is not full ASL grammar translation and does not replace interpreters.
WHY we used it	Honest limits make the project more credible and protect the team from overclaiming.
WHY NOT the other approach	Claiming full ASL translation would be easy to challenge and hard to prove.
HOW it works	Write limitations clearly: controlled vocabulary, lighting/camera effects, prototype avatar, phrase disabled, training in Python.
WHY it is relevant	Relevant because oral defense depends on alignment between claims and evidence.
WHAT is irrelevant or delayed	Overclaiming is irrelevant and dangerous.
CHAPTER 1-3 connection	Chapter 1 scope/limitations; Chapter 3 ethics and summary.

6. Why This Approach, Not This-and-That: Quick Comparison Table

Option	Use now?	Why or why not
Raw CNN image-only recognition	No, not as final main pipeline	Good starting point, but live camera conditions caused problems. Keep as history and RRL support.
Landmark-based recognition	Yes	More controllable and less affected by background, lighting, and crop problems.

Static 63-feature hand model	Yes for alphabet	Efficient and suitable for A-Z, del, space, and nothing.
30x162 one-hand dynamic model	Yes for protected baseline	Captures motion and body-relative context while keeping one-hand accessibility.
FullSign225	Experimental	Potentially better for signs needing both hands, but must stay separate until proven.
TCN	Yes, current main dynamic direction	Practical for fixed 30-frame sequences and lightweight enough for TFLite direction.
LSTM	Comparison/back-up	Useful sequence model, but not automatically better for current live behavior.
Residual Dilated TCN	Experimental	May improve wider motion understanding but must not overwrite baseline.
Transformer/ST-GCN	Future only	Too complex/data-hungry for immediate defense-ready hardening.
Phrase-intent shortcuts	Frozen/optional	Promising but must wait until basics are reliable and endpoint segmentation is mature.
50+ vocabulary expansion	Not now	More words can increase confusion before demo10 and NOTHING are stable.
Cloud recognition	Not now	Conflicts with privacy/offline/low-latency goal.
Gloves/sensors	No	Adds hardware cost and reduces accessibility.
Full ASL grammar translation	No	Too broad for current capstone stage; current system is selected sign recognition.
FSL support	Future	Current dataset and scope are ASL-oriented; FSL can be added later with separate data and labels.

7. Paper Holes, Gaps, and How to Plug Them

A paper hole means a weak spot where the panel may ask: Why did you do that? How does this connect? Is this claim true? The table below lists the holes and the correct defense-ready fix.

Possible hole in the paper	Why it is a problem	How to fix/explain it
Paper says CNN is the main model.	The current system uses landmarks and temporal models.	Say CNN was an early approach; current pipeline is landmark-based with TCN/LSTM dynamic recognition.
Paper says static alphabet only.	Current system includes selected dynamic words and NOTHING.	Update Chapter 1 objectives and Chapter 3 methodology to include static + dynamic + no-output class.
No clear reason for Android.	The system is meant to be portable and usable in daily places.	Explain Android as the target local device platform using CameraX, MediaPipe, and TFLite.
No explanation for offline/on-device.	Accessibility may fail if internet is required.	Explain privacy, latency, and network-limited environments.
No explanation for one-hand/dominant hand.	Panel may ask why not full two-hand ASL.	Explain accessibility for one-hand users and controlled reliability for current stage.
No explanation for FullSign225.	Panel may think the team is changing architecture again.	Explain it as a separate experiment, not a replacement until proven.
No explanation for NOTHING.	False positives are a major live risk.	Explain NOTHING as a no-output class for idle, partial, transition, and non-sign movement.
No gates in methodology.	Raw predictions are unsafe for communication.	Add confidence, margin, motion, wrist path, hand presence, quality, stability, and cooldown.
No token composer explanation.	It may look like the model directly creates sentences.	Explain that accepted tokens build text; this avoids claiming full ASL grammar translation.
Phrase-intent confusion.	ASK_NAME could be misunderstood as full sentence translation.	Say phrase-intent exists as a shortcut/future path but is currently disabled during hardening.
Avatar overclaim.	A simple avatar is not full ASL grammar.	Call it lightweight visual feedback/fingerspelling, not a complete ASL avatar interpreter.

Chapter 2 RRL too generic.	Studies must connect to system choices.	Group RRL by sign recognition, CNN, landmarks, temporal models, edge Android, avatar/STT, ISO/IEC 25010.
Chapter 3 too vague.	Methodology must describe what was actually built.	Include research design, Agile RAD, architecture, datasets, features, models, gates, Android, avatar, testing, evaluation.
References unrelated to system.	Irrelevant sources weaken credibility.	Remove unrelated template references and use sign language recognition, MediaPipe, TCN/LSTM, TFLite, edge AI, ISO/IEC 25010.
No ethics section.	Camera and user recording involve privacy.	Add consent, privacy, careful handling of recordings, and honest prototype limitations.
No current/future scope separation.	Panel may think unfinished parts are failed promises.	Separate current defense-ready scope from future/experimental paths.

8. Chapter 1 Expansion Logic

Chapter 1 should read like the reason for the whole project. It should not sound like the team randomly chose AI. It should show a straight path from problem to solution.

Chapter 1 part	What it should say	Why it matters
Background	Communication barriers happen when signing users meet hearing/non-signing people without interpreter support.	This proves the problem is real and human-centered.
Project Context	VoxGest uses AI, computer vision, Android, and accessibility technology to build a local assistive app.	This places the project in BSIT/AI/mobile development context.
Purpose	Recognize selected signs, convert accepted signs to text/speech, and support replies with STT/avatar/fingerspelling.	This explains bidirectional communication.
Statement of the Problem	Ask how to design Android app, extract landmarks, recognize static/dynamic signs, reduce false output, and evaluate quality.	Each question maps to a system component.
Objectives	Build Android prototype, implement landmarks, static model, dynamic model, NOTHING/gates, token composer, TTS/STT/avatar, ISO evaluation.	Objectives become measurable deliverables.
Scope and Limitation	Controlled vocabulary, not full ASL, not interpreter replacement, phrase disabled, FullSign225 experimental, Android inference only.	This protects the team from overclaiming.

9. Chapter 2 Expansion Logic

Chapter 2 should not be a random pile of studies. Every literature section must support a decision in VoxGest.

RRL category	Why it belongs	What it supports in VoxGest
Sign language recognition and assistive communication	Shows the communication problem and why recognition systems matter.	Chapter 1 problem and project purpose.
CNN and deep learning	Supports the original image recognition attempt and early model direction.	CNN history and broader AI basis.
Landmark tracking and MediaPipe	Supports the shift from images to hand/body coordinates.	Feature extraction and Python/Android preprocessing.
Dynamic recognition using LSTM/TCN	Supports moving sign recognition across frames.	30-frame word model and TCN/LSTM choice.
Negative class and hardening	Supports why models need no-output classes and rejection rules.	NOTHING, gates, reliability.
Android edge computing and TFLite	Supports offline, local, mobile recognition.	Android deployment and runtime contract.
Speech-to-text, TTS, and avatar feedback	Supports bidirectional communication.	Reverse communication flow and visual response.
ISO/IEC 25010	Supports formal software quality evaluation.	Chapter 3 evaluation and future Chapter 4 results.

RRL rule:

If a study does not help explain one of the system choices, it is weak for this paper. The literature must defend the system, not decorate the paper.

10. Chapter 3 Expansion Logic

Chapter 3 should explain the actual build path. It should answer what was built, why it was built that way, and how each part works.

Methodology part	What to include	Simple reason
Research Design	Developmental research.	The output is a working technology artifact.
Development Methodology	Agile/RAD build-test-revise cycles.	The system changed through live testing and repair.
System Architecture	Camera -> landmarks -> models -> gates -> token composer -> outputs.	Shows the complete logic chain.
Dataset Development	Static signs, dynamic sequences, NOTHING negatives, teammate recorder workflow.	Shows how training examples were prepared.
Feature Engineering	63 static, 30x162 dynamic, FullSign225 30x225, optional phrase 60x162.	Explains model input numbers.
Model Development	CNN -> hybrid -> landmark static -> LSTM/TCN -> RD-TCN experiment.	Shows honest evolution instead of hiding failed attempts.
Recognition Hardening	NOTHING, gates, live logger, dataset audit, cooldowns.	Shows reliability and safety.
Android Development	Kotlin, CameraX, MediaPipe, TFLite, same preprocessing as Python.	Connects prototype to actual app deployment.
Avatar/STT/TTS	TTS for signed output, STT for reply, avatar/fingerspelling for visual response.	Completes bidirectional communication.
Testing Procedure	Static test, dynamic validation, live logs, negative class test, Android readiness.	Gives evidence path.
Evaluation	ISO/IEC 25010 with functional suitability, performance efficiency, reliability, usability.	Provides formal software evaluation.

11. Relevance and Irrelevance Map

Some approaches are relevant now. Some are not wrong, but they are irrelevant to the current defenseready stage. This section helps the team explain the difference.

Approach/idea	Relevant now?	Reason
Static alphabet recognition	Yes	It is the spelling and control foundation.
demo10 dynamic words + NOTHING	Yes	It is the protected baseline for current defense/demo.
Token composer	Yes	It builds safe text from accepted predictions.
Recognition gates	Yes	They prevent weak predictions from becoming output.
Android/TFLite	Yes	It is the target deployment direction.
Lightweight avatar/fingerspelling	Yes	It supports the reverse visual communication path.
FullSign225	Experimental	Relevant as research direction, but not default until proven.
Residual Dilated TCN	Experimental	Relevant as separate model improvement, but not a replacement yet.
Phrase intents	Frozen	Relevant as future path, but disabled because basics must be stable first.
50-word expansion	Not now	More words before hardening may increase confusion.
Full ASL grammar	Not now	Too broad and not supported by current model scope.
Interpreter replacement claim	No	Unethical/overclaiming; VoxGest is an assistive prototype.

Cloud recognition	No for current system	Conflicts with offline/privacy/local goal.
Hardware gloves/sensors	No	Would reduce low-cost phone-based accessibility.

12. The Strongest Oral Defense Logic

Defense answer:

VoxGest started with CNN because image recognition was a logical first step. Live testing showed that raw images were fragile in real camera conditions, so the system moved to landmarks. Static letters use 63 hand features. Dynamic words use 30-frame landmark sequences with body context. TCN is the current main dynamic direction, LSTM is comparison/back-up, and FullSign225/RD-TCN are experimental paths kept separate from the protected baseline. NOTHING, gates, and token composer exist so the app does not speak or animate unsafe raw predictions.

Common Panel Questions and Safe Answers

Question	Safe answer
Is VoxGest a full ASL translator?	No. It is a controlled sign-recognition and communication prototype. It supports static signs, selected dynamic words, token-based output, and lightweight visual feedback.
Why did you not stay with CNN?	CNN was logical for early image classification, but live camera testing showed background, lighting, cropping, and camera-angle problems. Landmarks are more controllable for live Android use.
Why use landmarks?	Landmarks turn the hand/body into numerical points. This helps the model focus on geometry instead of the whole room.
Why do dynamic signs need pose?	Some signs depend on where the hand is near the body. Pose gives body-location context.
Why is NOTHING important?	It prevents idle hands, partial signs, or transitions from becoming fake words.
Why freeze phrase recognition?	Phrase recognition is promising, but the current system must first make alphabet, 10 words, NOTHING, and token composition reliable.
Why one hand?	It supports accessibility for users who rely on one hand and keeps the baseline simpler for real-time mobile use.
Why FullSign225?	It is a separate experiment for signs needing both hands/body information. It is not promoted until live testing proves it.
Why Android?	Android provides portable camera-based deployment, speech tools, and TensorFlow Lite for on-device inference.
How will you evaluate it?	Using ISO/IEC 25010: functional suitability, performance efficiency, reliability, and usability.

13. Final System Logic Chain

This is the shortest complete chain. Memorize this for defense:

15. The user problem is communication between signing users and hearing/non-signing people.
16. The product goal is an offline Android app that helps both sides communicate.
17. The camera captures hand/body movement.
18. MediaPipe/Holistic converts the video into landmark dots.
19. Static model recognizes A-Z, del, space, and nothing from 63 hand features.
20. Dynamic model recognizes selected words from 30-frame temporal landmark sequences.
21. NOTHING and gates block idle, weak, partial, or unsafe predictions.
22. The token composer builds text only from accepted signs.
23. Text-to-speech lets the hearing person hear the signed message.
24. Speech-to-text lets the signing user read the hearing person reply.
25. Avatar/fingerspelling gives a simple visual response.
26. ISO/IEC 25010 evaluates whether the system is useful, fast, reliable, and usable.

Final team rule:

Do not present future experiments as finished features. Present demo10 and Recognition Hardening v1 as the current safe core. Present FullSign225, RD-TCN, and phrase intents as controlled experimental/future paths.

14. Source Basis Used

This companion guide was based on the current VoxGest project documentation and paper alignment files, including the comprehensive contextual notes, simplified contextual notes, Chapter 1-3 updated draft, Android Developer Handoff, Current Demo System/README, Progress Log, ML Engineer Knowledge Base, Recording Protocol, Avatar Pipeline, and Sentence Intent notes.